

Router
Switch
Firewall
AP

Local configuration
→ Server network

Global configuration
→ Cisco

Number System

- Binary (0-1)
- Decimal (0-9)
- Hexa Decimal 0-9
10 - A
11 - B
12 - C
13 - D
14 - E
15 - F

Base value :-

Total numbers in a number system
is called Base Value

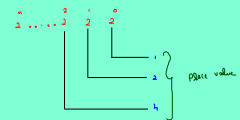
Binary = 2

Decimal = 10

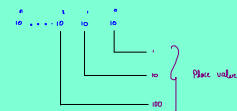
Hexa = 16

Place Value :-

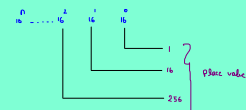
Binary :-



Decimal



Hexa



conversion

Decimal → Binary

1 octet = 8 bits

128	64	32	16	8	4	2	1
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128	64	32	16	8	4	2	1
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10 0 0 0 0 1 0 1 0

20 0 0 0 1 0 1 0 0

192 1 1 0 0 0 0 0 0

240 1 1 1 1 0 0 0 0

255 1 1 1 1 1 1 1 1

1 → come

0 → don't come

Binary to Decimal :

128	64	32	16	8	4	2	1
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1 1 0 0 1 0 1 1 = 203

0 0 1 1 1 0 1 1 = 59

1 1 0 1 1 1 1 1 = 223

1 1 1 0 1 0 1 0 = 234

1 1 1 1 0 1 1 0 = 246

Hexa to Binary

1 Hexa = 4 bits

8	4	2	1
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8	4	2	1
---	---	---	---

10 A 1 0 1 0

11 B 1 0 1 1

12 C 1 1 0 0

13 D 1 1 0 1

14 E 1 1 1 0

15 F 1 1 1 1